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A NOTE ON FOURIER SERIES OF FUNCTIONS BELONGING TO $Lip_M(\alpha, A)$, $A = \{C, L_1\}$

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Abstract

An estimation of Fourier coefficients of functions belonging to $Lip_M(\alpha, C)$ class is provided.

1. Introduction and Auxiliary Facts

The fundamental problem in approximation theory is to determine the properties of the approximating characteristics of the function based on the axiomatic properties of the function, namely the relation between the structural characteristics of a function (its differentiation, the properties of its modulus of continuity, the upper bound of the function, satisfying the Lipschitz condition, etc.) and the constructive characteristics of that function (the zero velocity of the range of the best approximations with its algebraic or trigonometric polynomials, the convergence of the Fourier series, the approximation by polynomials generated by linear methods of the Fourier series summation, the approximation by interpolating polynomials, etc.).

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So in this paper is given an estimation of Fourier coefficients of functions belonging to $f \in Lip_M(\alpha, C)$ and $f \in Lip_M(\alpha, L_1)$ class.

Denote by L_p the set of 2π-periodic functions f, such that f is measurable on $[0, 2\pi]$ for $p \in [1, \infty)$ and f is continuous on $[0, 2\pi]$ for $p = \infty$, respectively

$$\| f \|_{L_p} = \begin{cases} \left(\int_0^{2\pi} | f(x) |^p dx \right)^{1/p} < \infty, & \text{for } p \in [1, \infty), \\ \max\limits_x | f(x) |, & \text{for } p = \infty. \end{cases} \tag{1.1}$$

For $p = \infty$, L_∞ space is denoted by C and represents the set of the continuous functions.

Denote by $\omega(f, t)$ the modulus of continuity of the function f belonging to C, respectively

$$\omega(f, t) = \max_{\substack{h \in [-t, t] \\ x \in [0, 2\pi]}} | f(x + h) - f(x) |. \tag{1.2}$$

Denote by $\omega_1(f, t)_1$ the modulus of continuity of the function f belonging to L_1, respectively

$$\omega_1(f, t)_1 = \sup_{|t| < h} \| \Delta_t^1 f \|_1, \quad \Delta_t^1 f(x) = f\left(x + \frac{t}{2}\right) - f\left(x - \frac{t}{2}\right). \tag{1.3}$$

Definition 1.1. By $L(0, 2\pi)$ denote the set of 2π-periodic functions summable on $(0, 2\pi)$. Let $f \in L(0, 2\pi)$ and let $a_m = a_m(f)$ and $b_m = b_m(f)$, $m = 0, 1, 2, \ldots$ be the Fourier coefficients of f, respectively

$$S[f] = \frac{a_0}{2} + \sum_{m=1}^{\infty} (a_m \cos mx + b_m \sin mx). \tag{1.4}$$

Note 1.1. For any $p, q, 1 < p < q < \infty$ it is clear that $C = L_\infty \subset L_q \subset L_p \subset L$.

Let $\alpha, M > 0$ and $f \in C$. We say that the function $f \in C$ satisfies a Lipschitz condition of order α related to the constant M in spaces C if $\omega(f, t) \le M|t|^{\alpha}$ and symbolically is written that $f \in Lip_M(\alpha, C)$.

Proposition 1.1. *Let* $\frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos kx + b_k \sin kx$ *be the Fourier series of the function* $f(x) \in L_1$. *Then* $|a_k|, |b_k| \le \frac{1}{\pi}\|f\|_{L_1}$.

2. Main Results

Theorem 2.1. *Let* $\frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos kx + b_k \sin kx$ *be the Fourier series of the function* $f \in Lip_M(\alpha, C)$. *Then* $\sum_{k=n}^{2n-1} |a_k b_k| \le Mn^{-2\alpha}$.

Proof. Let $g(x) = f(x+t) - f(x-t)$, where $f \in Lip_M(\alpha, C)$. Then

$$\frac{1}{\pi}\int_{-\pi}^{\pi} g^2(x)dx = \frac{1}{\pi}\int_{-\pi}^{\pi} (f(u+t) - f(u-t))^2 du$$

$$\le \frac{1}{\pi}\int_{-\pi}^{\pi} M_1(|2t|^{\alpha})^2 du = 4\frac{M_1}{\pi}|t|^{2\alpha}2\pi = 8M_1|t|^{2\alpha}.$$

The Fourier series of function $g(x)$ is $2\sum_{k=1}^{\infty}(-a_k \sin kx + b_k \sin kx)\sin kt$.

Then using Parseval's equality we obtain that

$$\sum_{k=1}^{\infty}((a_k \sin kt)^2 + (b_k \sin kt)^2) \le 2M_1|t|^{2\alpha}.$$

For any n and taking $t = \frac{\pi}{4n}$, since $\frac{1}{2} \le \sin^2 kt \le 1$, $k \in [n, 2n-1]$, we get:

$$\sum_{k=n}^{2n-1} | a_k b_k | \leq \sum_{k=n}^{2n-1} \frac{a_k^2 + b_k^2}{2} \leq \sum_{k=1}^{2n-1} ((a_k \sin kt)^2 + (b_k \sin kt)^2)$$

$$\leq 2M_1 | t |^{2\alpha} = 2M_1 \left| \frac{\pi}{4n} \right|^{2\alpha} = Mn^{-2\alpha}.$$

Theorem 3.2. *Let* $\frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos kx + b_k \sin kx$ *be the Fourier series of the function* $f \in L_M(\alpha, L_1)$, $\alpha > 1$. *Then* $f = c$, $c \in R$.

Proof. According to Proposition 1.1 we get the inequality

$$| a_k(\Delta_t^1 f) | = \frac{1}{\pi} \left| \int_{-\pi}^{\pi} \left(f\left(x + \frac{t}{2} \right) - f\left(x - \frac{t}{2} \right) \right) \cos kx dx \right|$$

$$= \frac{1}{\pi} \left| \int_{-\pi}^{\pi} \left(- \cos k\left(x + \frac{t}{2} \right) + \cos k\left(x - \frac{t}{2} \right) \right) f(x) dx \right|$$

$$= \left| 2 \sin \frac{kt}{2} b_k(f) \right| \leq \frac{1}{\pi} \| \Delta_t^1 f \|_1$$

$$\leq \frac{1}{\pi} \sup_{-h<t<h} \| \Delta_t^1 f \|_1 = \frac{\omega_1(f, h)_1}{\pi}.$$

Respectively $|b_k(f)| \leq \frac{\omega_1(f, h)_1}{2\left|\sin \frac{kt}{2}\right| \pi}$. Setting $t = t_i$, $i \in N$, such that $t_i > 0$ and $\lim_{i \to \infty} t_i = 0$, and taking limit both sides we get

$$\lim_{i\to\infty} \left| b_k(f) \right| \leq \lim_{i\to\infty} \frac{\frac{kt_i}{2} \omega_1(f, h_i)_1}{2\left| \sin \frac{kt_i}{2} \right| \pi \frac{kt_i}{2}} = \frac{1}{k\pi} \lim_{i\to\infty} \frac{\omega_1(f, h_i)_1}{t_i}$$

$$\leq \frac{1}{k\pi} \lim_{i\to\infty} M \frac{h_i^{\alpha}}{t_i} \leq \frac{1}{k\pi} \lim_{i\to\infty} M t_i^{\alpha-1} = \frac{1}{k\pi} 0 = 0 \Rightarrow b_k(f) = 0, \ \forall k \in N.$$

Similarly we find $a_k(f) = 0, \forall k \in N$.

So we get $S[f] = \frac{a_0}{2} = c \in R \Rightarrow f(x) = c$.

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